

## Questions by topic: Biogeochemical cycles: carbon, nitrogen, phosphorus - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1

Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

**Table 1**

| Reservoir                     | Mass of carbon / billion tonnes |
|-------------------------------|---------------------------------|
| Atmosphere (CO <sub>2</sub> ) | 700                             |
| Living organisms              | 470                             |
| Dead organic matter (soil)    | 3 700                           |
| Surface ocean waters          | 500                             |
| Deep ocean (dissolved)        | 35 000                          |
| Fossil fuels                  | 10 000                          |
| Sedimentary carbonate rocks   | 20 000 000                      |

0 1

. 1

Express the mass of carbon in the atmosphere in standard form, in tonnes.

**[1 mark]**

Answer

tonnes

0 1

. 2

Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

**[2 marks]**

Answer

%

0 1

. 3

Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO<sub>2</sub> concentrations.

**[2 marks]**

.....

01 . 4

Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

01 . 5

Explain why aerobic respiration returns carbon to the atmosphere as  $\text{CO}_2$  but anaerobic respiration returns it mainly as methane.

[2 marks]

10

02 . 1

Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

**Ploughing farmland** .....

.....

.....

.....

**Draining marshland / peat bogs** .....

.....

.....

.....

**Toxic pollution of the sea** .....

.....

.....

.....

**0 2 . 2**

Give **three** anaerobic environments created by human activities that release methane.

**[3 marks]**

**1** .....

.....

**2** .....

.....

**3** .....

.....

**0 2 . 3**

Explain how rising atmospheric CO<sub>2</sub> causes ocean acidification.

**[1 mark]**

.....

.....

**10**

03 . 1

Explain why planting a new forest only sequesters carbon for a limited period, and how the storage can be extended.

**[3 marks]**

.....

.....

.....

.....

.....

.....

03 . 2

Compare pre-combustion and post-combustion methods of carbon capture.

**[4 marks]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

03 . 3

Describe **two** ways captured CO<sub>2</sub> can be stored and give **one** concern about long-term storage.

**[3 marks]**

.....

.....

.....

.....

.....

.....

10

0 4

Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

Figure 1

| Number | Process description                                                            |
|--------|--------------------------------------------------------------------------------|
| 1      | Nitrogen gas converted to ammonium ions by Rhizobium in root nodules           |
| 2      | Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter |
| 3      | Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil              |
| 4      | Amino groups in dead organic matter released as ammonium ions by decomposers   |
| 5      | Nitrogen and oxygen react during lightning to form NO <sub>x</sub>             |
| 6      | Nitrates washed out of soil into rivers                                        |

0 4

1

Name processes 1 to 6.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4

2

Explain why draining a waterlogged field increases the nitrate available to crops.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 3

Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

[2 marks]

.....

.....

.....

0 4 . 4

Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

[2 marks]

.....

.....

.....

10

0 5

Table 1 shows global reactive nitrogen fixed each year by natural processes and by human activities.

Table 1

| Source                                   | Nitrogen fixed / million tonnes per year |
|------------------------------------------|------------------------------------------|
| Biological fixation (natural ecosystems) | 110                                      |
| Lightning                                | 5                                        |
| Haber process (fertiliser manufacture)   | 120                                      |
| Cultivation of legume crops              | 60                                       |
| Combustion (NO <sub>x</sub> )            | 30                                       |

0 5 . 1

Use Table 1 to calculate the percentage of total nitrogen fixation that is caused by human activities.

[2 marks]

Answer

%

05 . 2

Describe the Haber process and explain why it is unsustainable compared with biological nitrogen fixation.

[3 marks]

.....

.....

.....

.....

.....

.....

05 . 3

Describe **three** consequences of the increased movement of nitrogen compounds caused by human activities.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

05 . 4

Explain how composting or anaerobic digestion of biological wastes produces a better fertiliser than raw waste.

[2 marks]

.....

.....

.....

.....

10

06 . 1

Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

**[3 marks]**

.....

.....

.....

.....

.....

.....

06 . 2

Explain how mycorrhizal fungi increase phosphate uptake by plants.

**[2 marks]**

.....

.....

.....

.....

06 . 3

Describe how human activities have altered the phosphorus cycle.

**[3 marks]**

.....

.....

.....

.....

.....

.....

06 . 4

Explain **two** strategies for the sustainable management of phosphorus supplies.

**[2 marks]**

.....

.....

.....

.....

10

|   |   |   |   |
|---|---|---|---|
| 0 | 7 | . | 1 |
|---|---|---|---|

Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

**[9 marks]**

Extra space

**07 . 2** State what is meant by the **residence time** of a substance in a reservoir. **[1 mark]**

**10**

**0 8**

A student measured the mass of carbon stored per hectare in three land uses in the same region.

**Table 1**

| Land use                   | Carbon in vegetation / t ha <sup>-1</sup> | Carbon in soil dead organic matter / t ha <sup>-1</sup> |
|----------------------------|-------------------------------------------|---------------------------------------------------------|
| Mature woodland            | 120                                       | 95                                                      |
| Permanent pasture          | 3                                         | 80                                                      |
| Arable (ploughed annually) | 2                                         | 45                                                      |

**0 8****1**

Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

**[1 mark]**

Answer

t/ha

**0 8****2**

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

**[2 marks]**

.....

.....

.....

.....

**0 8****3**

Suggest **two** ways the arable farmer could increase soil carbon storage.

**[2 marks]**

**1** .....

.....

**2** .....

.....

**5****END OF QUESTIONS**